Cover census over cubic extensions: the minimal GHS genus is set by #E(K) mod 4

2026-10-12. Conductor T-2261. **Structural stage diagnostic on toy fields.** It counts covers by Riemann–Hurwitz genus. No curve over F_q is constructed, no transfer map is evaluated, no discrete logarithm is solved, and no speedup is claimed. Protocol frozen before the first run: PROTOCOL.md. Run 20261013T0305Z: records, environment, run report. This report as PDF: REPORT.pdf. Follows research/weak_subfield_hyperelliptic_decomposition_20261012/ (PR #1757), which showed that same-field higher-genus representations do not help.

Question

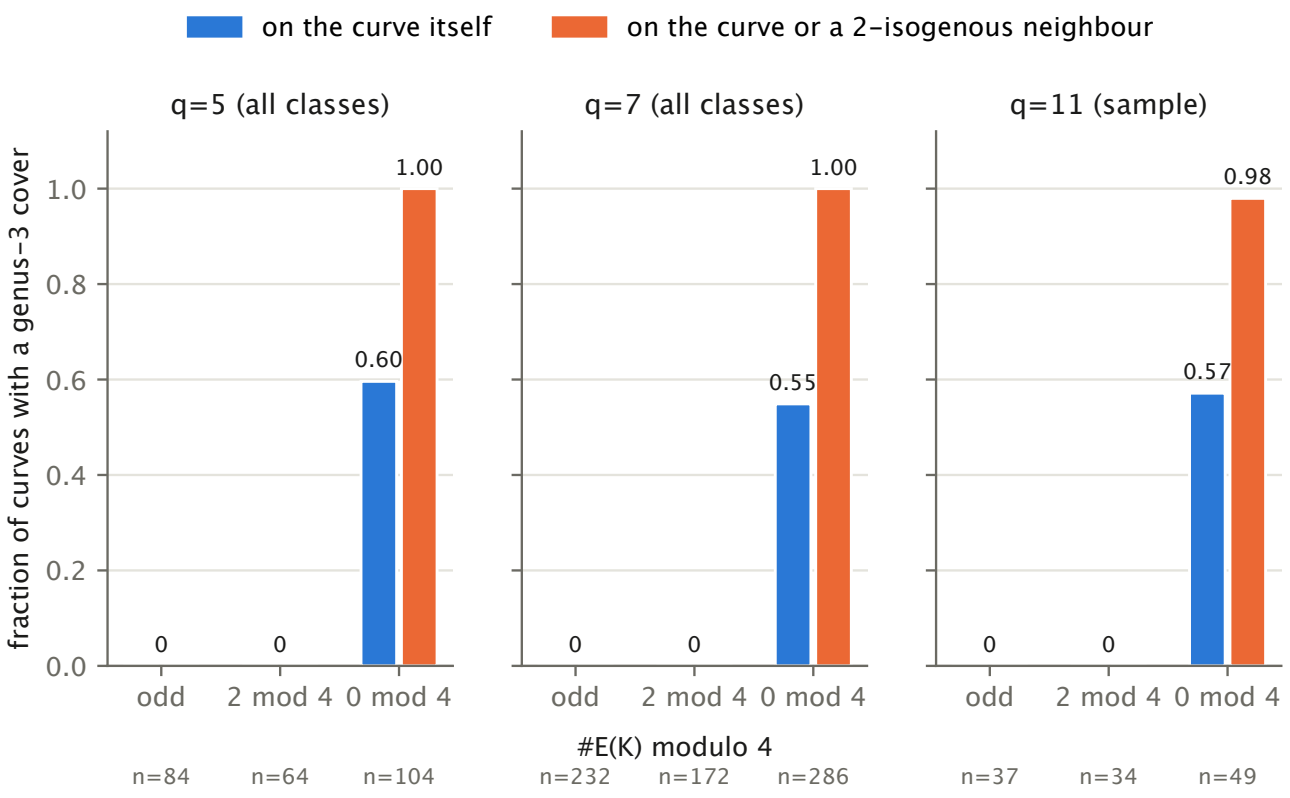
For E over K = F_{q³}, what is the smallest genus of a curve over F_q that the odd-characteristic GHS construction attaches to E, over every F_q-structure on the x-line? What changes after a 2-isogeny, and over the whole isogeny class?

Result

With one exception the minimal genus over the isogeny class follows #E(K) mod 4 (counts are isomorphism classes of curves):

#E(K)	minimal genus over the isogeny class	status
odd	13	derived (below) and measured, 316 of 316 curves at q = 5, 7 and 37 of 37 sampled at q = 11
2 mod 4	5 (one exception: 9)	measured: 5 on 234 of 236 curves at q = 5, 7, 9 on the other two (one class at q = 5); never 3
0 mod 4	3	measured, 390 of 390 curves at q = 5, 7

Genus 3 is the only value below the rho exponent (PROTOCOL.md), so on these fields a cover that could matter exists exactly for the classes with 4 | #E(K), and a curve of prime order, or of order twice an odd number, has none. The "2 mod 4" and "0 mod 4" rows are observations on q ∈ {5, 7} (and a sample at q = 11), not theorems.



Measurements (run 20261013T0305Z, exact counts)

Tool: cover_census.rs. For each curve it runs through every cocycle A (the tool found exactly (Q³−Q)/(q³−q) of them in each field: 16275, 120099, 1786323). The genus formula agreed with the sum of the genera of the quadratic subcovers on every structure where it was checked (genus at most 5, or at the running minimum): 0 mismatches.

Curves by minimal genus, split by #E(K) mod 4. "self" is the curve, "≤1 step" the minimum over the curve and its K-rational 2-isogeny neighbours, "class" the minimum over all curves with the same #E(K).

field	#E(K)	curves	self: genus 3 / 5 / 9 / 13	≤1 step: genus 3 / 5 / 9 / 13	class: genus 3 / 5 / 9 / 13
q = 5, all 252 classes	odd	84	0 / 0 / 0 / 84	0 / 0 / 0 / 84	0 / 0 / 0 / 84
	2 mod 4	64	0 / 60 / 4 / 0	0 / 60 / 4 / 0	0 / 62 / 2 / 0

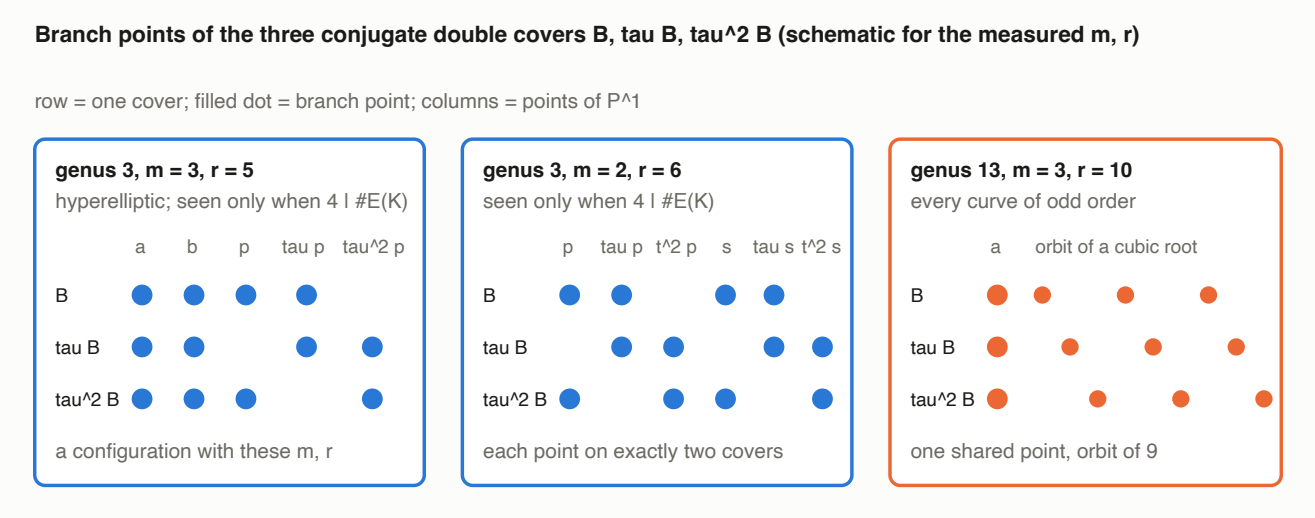
field	#E(K)	curves	self: genus 3 / 5 / 9 / 13	≤1 step: genus 3 / 5 / 9 / 13	class: genus 3 / 5 / 9 / 13
	0 mod 4	104	62 / 42 / 0 / 0	104 / 0 / 0 / 0	104 / 0 / 0 / 0
q = 7, all 690 classes	odd	232	0 / 0 / 0 / 232	0 / 0 / 0 / 232	0 / 0 / 0 / 232
	2 mod 4	172	0 / 168 / 4 / 0	0 / 168 / 4 / 0	0 / 172 / 0 / 0
	0 mod 4	286	157 / 126 / 3 / 0	286 / 0 / 0 / 0	286 / 0 / 0 / 0
q = 11, sample of 120	odd	37	0 / 0 / 0 / 37	0 / 0 / 0 / 37	not computed
	2 mod 4	34	0 / 33 / 1 / 0	0 / 33 / 1 / 0	not computed
	0 mod 4	49	28 / 21 / 0 / 0	48 / 1 / 0 / 0	not computed

At `q = 5` two curves of order 2 mod 4 sit in an isogeny class whose minimum is 9; at `q = 7` every such class reaches 5.

Types at the minimum (number of curves; a curve can have both genus-3 types):

field	genus 3, `m=2, r=6` only	both	genus 3, `m=3, r=5` only (hyperelliptic)	genus 5, `m=3, r=6` (hyperelliptic)	genus 9, `m=3, r=8`	genus 13, `m=3, r=10`
q = 5	18	24	20	102	4	84
q = 7	66	49	42	294	7	232
q = 11 (sample)	18	4	6	54	1	37

A genus-3 curve has between 2 and 18 structures attaining it at `q = 5` and between 2 and 24 at `q = 7`, out of 16275 and 120099. The smallest cofactor among curves with a genus-3 cover is 4 (`#E(K) = 124 = 4·31` at `q = 5`, `332 = 4·83` at `q = 7`).



Derivation for odd order

Proposition (derived; orbit step in Lean). If `#E(K)` is odd, every structure gives genus at least 13, and 13 is attained.

Proof. The branch set is one `K`-rational point `a` and the three roots of a cubic irreducible over `K`. A root `θ` has `τ⁹θ = θ` and `τ³θ ≠ θ` (`τ³` is the Frobenius of `K`), so `τ^kθ ≠ θ` for `1 ≤ k ≤ 8` (`OrbitNine.no_short_return`) and the nine iterates are distinct. They are all branch points of some `τ^i B`, so `r ≥ 9 + 1`, with equality iff `τa = a`. No two of `B, τB, τ²B` coincide and their triple symmetric difference contains `θ`, so `m = 3` and `g = 2(r − 4) + 1 ≥ 13`. ∎

The same holds for every curve isogenous over `K`, since the order is an isogeny invariant. P2 of the protocol predicted "at least 5"; the proof and the data give 13.

Lean: lean/OrbitNine.lean, toolchain v4.34.1, no Mathlib; `#print axioms` reports `propext` only.

Report statement	Lean declaration	Status
no return of `θ` before 9 steps	`no_short_return`	proved
`τ³` is the Frobenius of `K`; branch set; genus formula	none	**not formalized**

No identity certificate: the report claims no polynomial identity.

What this does and does not say

- **Prime-order curves over a cubic extension have no small cover of this kind**, and no isogeny changes that. The route needs cofactor at least 4.

- **For `4 | #E(K)` a genus-3 cover is one 2-isogeny away** on 438 of 439 curves examined (104 + 286 + 48); the remaining `q = 11` curve was not followed further.
- **Not established:** that the genus-3 curve has a model over `F_q` with the map `E(K) → Jac(C)(F_q)` injective on the odd part (standard for odd `n`, cited from memory, not verified here); whether the `m = 2` curves are plane quartics; any cost. The exponents in PROTOCOL.md are published asymptotics, not measurements.
- **Scope:** `n = 3`, `q ∈ {5, 7, 11}`, odd characteristic above 3, the `(2,…,2)` construction only. Other covers (trigonal, `n = 5, 7`, characteristic 2) are not covered. A bounded census is not a proof for all `q`.

Next discriminating step

Build the genus-3 curve over `F_q` for one `q = 7` class of order `4·ℓ` and check the transfer on the order-`ℓ` subgroup. If it is injective, a genus-3 index-calculus count against rho on that toy instance is the first measurement that could say anything about cost. If the `F_q`-model fails to exist, the census counts something the attack cannot use.

Graphs checked and left unchanged

No canonical graph changes (no ECDLP rung, no timing; no `benchmarks/` entry).

Reproduce

```
rustc -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census 11 120
python3 reports/plot.py results/run-20261013T0305Z.jsonl reports/run-20261013T0305Z   # layout only
cd lean && lake build
```